FOSSEE OSHW Hackathon 2026 Report

On

Arduino Based Energy Optimization System

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Abstract

This project presents an Arduino-based energy optimization system designed to automatically control lighting based on human presence detection. The system utilizes a Passive Infrared (PIR) motion sensor to detect movement in a room and automatically controls two LED lights representing room lighting and outside lighting. A 16×2 LCD display provides real-time status updates about the system's operational mode. This automation solution helps reduce energy consumption by ensuring lights are used only when needed.

Introduction

In today's modern world, the demand for electrical energy is increasing rapidly due to the growing use of electrical appliances in homes, offices, industries, and public places. However, a large amount of electricity is wasted because lights, fans, and other devices are often left switched ON even when no one is present in the room. This unnecessary consumption of electricity not only increases electricity bills but also leads to inefficient use of energy resources. Therefore, it is very important to develop systems that can help in **optimizing energy usage and reducing wastage**.

Automation technology has become an effective solution for improving energy efficiency. With the advancement of **microcontrollers and sensors**, it is now possible to design smart systems that can automatically control electrical devices without human intervention. One of the most commonly used microcontroller platforms for such applications is the **Arduino Uno**, which is easy to program, low cost, and widely used in embedded system projects.

This project presents an **Arduino-Based Energy Optimization System with PIR Motion Control**, which aims to reduce electricity wastage by automatically controlling electrical appliances based on human presence. The system uses a **Passive Infrared Sensor (PIR sensor)** to detect motion in a particular area. When a person enters the room, the PIR sensor detects the infrared radiation emitted by the human body and sends a signal to the Arduino microcontroller. The Arduino then processes this signal and activates a relay module to turn ON the connected electrical appliance such as a light or fan. When no motion is detected for a certain period of time, the system automatically turns OFF the appliance, thereby saving energy.

The main goal of this project is to demonstrate how **simple automation techniques can significantly improve energy efficiency** in buildings and homes. This system is cost-effective, easy to implement, and can be used in various places such as classrooms, offices, corridors, parking areas, and smart homes. By implementing such systems on a larger scale, it is possible to reduce electricity consumption and contribute to energy conservation.

Thus, the Arduino-based energy optimization system provides a **smart, reliable, and efficient solution for automatic energy management**.

Objectives of the Project

The main objective of the project **"Arduino-Based Energy Optimization System with PIR Motion Control"** is to develop a smart and efficient system that reduces unnecessary electricity consumption by automatically controlling electrical appliances based on human presence. The project focuses on using simple automation technology to improve energy efficiency in homes, offices, and public spaces.

1. To Reduce Energy Wastage

One of the primary objectives of this project is to minimize the wastage of electrical energy. In many places, lights and fans remain ON even when no one is present in the room. By using a motion detection system, the project ensures that electrical appliances operate only when needed, thereby saving energy.

2. To Develop an Automatic Control System

The project aims to design an automatic system that can control electrical devices without human intervention.

Using a microcontroller like **Arduino Uno**, the system can automatically turn ON or OFF appliances depending on whether motion is detected in the room.

3. To Detect Human Motion Using Sensors

Another objective is to use a motion detection sensor to identify human presence. The system uses a **Passive Infrared Sensor (PIR sensor)** that detects changes in infrared radiation caused by human movement. This allows the system to activate electrical appliances only when someone is present.

4. To Create a Low-Cost Energy Saving System

The project also aims to develop a system that is affordable and easy to implement. Components like Arduino, PIR sensors, and relay modules are inexpensive and widely available, making the system suitable for practical applications.

5. To Promote Smart Energy Management

This project encourages the use of smart technology for efficient energy management. By integrating sensors and microcontrollers, the system demonstrates how

automation can help in reducing electricity consumption and improving overall energy efficiency.

6. To Provide Practical Knowledge of Embedded Systems

Another objective is to help students gain hands-on experience in embedded systems, sensor technology, and microcontroller programming. This project provides practical understanding of how hardware and software work together to build real-world automation systems.

Components Required

The **Arduino-Based Energy Optimization System with PIR Motion Control** uses several electronic components to detect human motion and control electrical appliances automatically.

1. Arduino Uno

Arduino Uno is the main microcontroller used in the project. It controls the whole system by receiving signals from the sensor and sending commands to the relay to switch appliances ON or OFF.

2. Passive Infrared Sensor (PIR Sensor)

The PIR sensor detects human motion by sensing infrared radiation emitted from the human body. When motion is detected, it sends a signal to the Arduino.

3. Relay Module and Resistors

The relay module acts as an automatic switch that allows the Arduino to control high-voltage electrical appliances like lights and fans.

4. LED or Bulb

An LED or bulb is used as the load in the project. It turns ON when motion is detected and turns OFF when no motion is present.

5. Breadboard

A breadboard is used to build and test the circuit without soldering. It helps in easily connecting different components.

6. Jumper Wires

Jumper wires are used to connect the Arduino, PIR sensor, relay module, and other components.

7. Power Supply

A power supply provides the required voltage to operate the Arduino and the entire system.

Component	Quantity	Purpose
Arduino UNO	1	Main microcontroller
PIR Motion Sensor	1	Detects human motion
16x2 LCD with I2C	1	Display system status
LEDs	2	Room and outside lights
Resistors 220Ω	2	Limit current
Breadboard	1	Connections
Jumper Wires	As needed	Interconnections

Working Principle

The **Arduino-Based Energy Optimization System with PIR Motion Control** works on the principle of **automatic detection of human presence and controlling electrical appliances accordingly**. The system mainly uses a motion sensor, a microcontroller, and a relay module to operate electrical devices such as lights or fans only when required. This helps in reducing unnecessary electricity consumption.

Step 1: Detection of Human Motion

The system uses a **Passive Infrared Sensor (PIR sensor)** to detect human movement. The PIR sensor works by sensing the **infrared radiation emitted by the human body**. When a person enters the detection area of the sensor, the infrared radiation pattern changes. The sensor detects this change and produces a digital signal.

Step 2: Signal Sent to the Microcontroller

The output signal from the PIR sensor is sent to the **Arduino Uno** microcontroller. The Arduino continuously monitors the input signal coming from the sensor. When the sensor detects motion, it sends a HIGH signal to the Arduino input pin.

Step 3: Processing of the Signal

After receiving the signal from the PIR sensor, the Arduino processes it according to the program stored in its memory. The program is written in the Arduino programming language and

uploaded through the Arduino IDE. The program instructs the Arduino to activate an output device when motion is detected.

Step 4: Activation of the Relay

The Arduino then sends an output signal to the **Relay Module**. The relay acts as an electrically operated switch that can control high-voltage electrical appliances using the low-voltage signal from the Arduino.

When the relay receives the signal from the Arduino:

- The relay switch closes.
- The electrical circuit connected to the appliance becomes complete.
- The appliance such as a light or fan turns ON.

Step 5: Automatic Switching OFF

When there is no movement in the area for a certain period of time, the PIR sensor stops sending signals to the Arduino. The Arduino then sends a command to the relay module to switch OFF the appliance. As a result, the electrical device turns OFF automatically, saving energy.

Circuit Diagram

The circuit of the **Arduino-Based Energy Optimization System with PIR Motion Control** consists of an **Arduino Uno**, a **Passive Infrared Sensor (PIR sensor)**, a **Relay Module**, an LED (or bulb), and a **resistor**. These components are connected together so that the system can detect human motion and automatically control electrical appliances.

1. PIR Sensor Connection

The PIR sensor has three main pins: **VCC, OUT, and GND**.

- **VCC** → Connected to the **5V pin of Arduino**
- **GND** → Connected to the **GND of Arduino**
- **OUT** → Connected to **Digital Pin 2 of Arduino**

The PIR sensor detects motion and sends a **HIGH signal (1)** to the Arduino when a person is present. When there is no motion, it sends a **LOW signal (0)**.

2. Relay Module Connection

The relay module is used to control high-voltage appliances.

- **VCC** → Connected to **5V of Arduino**
- **GND** → Connected to **GND of Arduino**
- **IN** → Connected to **Digital Pin 8 of Arduino**

The relay works like a switch. When the Arduino sends a signal to the relay, it turns the connected appliance **ON or OFF**.

3. LED and Resistor Connection

An LED is used to represent the electrical load in the circuit. A **220Ω resistor** is connected in series with the LED to limit the current and protect the LED from damage.

Connection steps:

- **Arduino Digital Pin 13 → Resistor (220Ω)**
- **Resistor → Anode (positive leg) of LED**
- **Cathode (negative leg) of LED → GND**

The resistor ensures that only a safe amount of current flows through the LED.

4. Appliance Connection through Relay

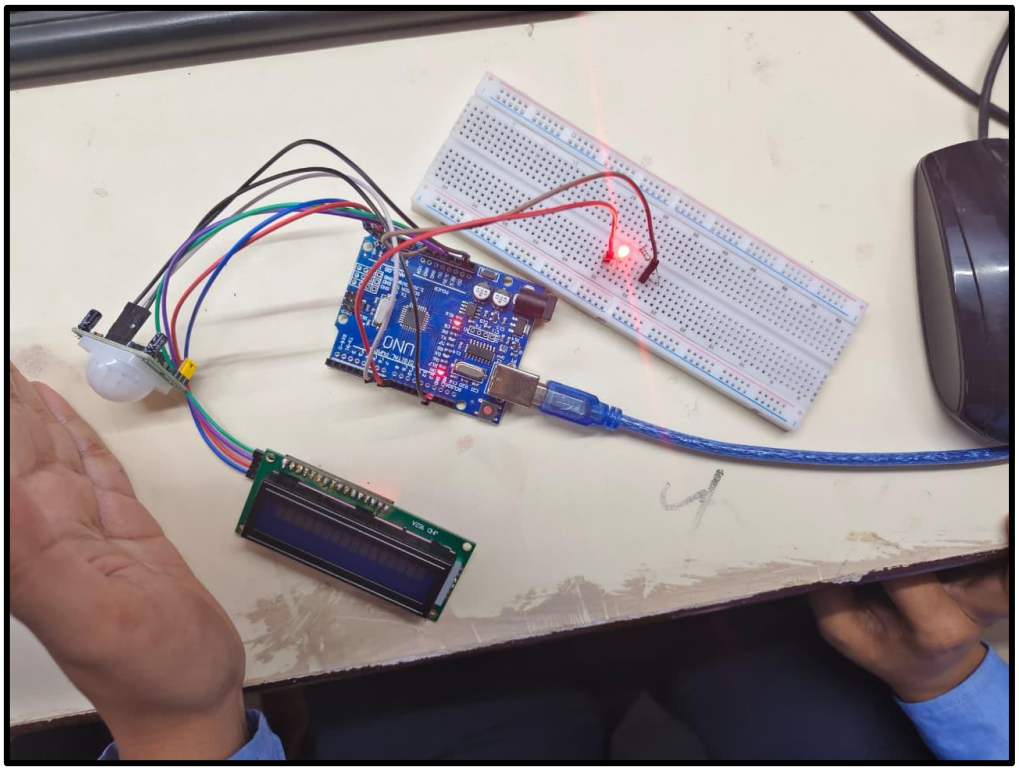
The electrical appliance (bulb or fan) is connected to the relay terminals.

- **COM (Common)** → Connected to power supply
- **NO (Normally Open)** → Connected to the appliance
- **Neutral wire** → Directly connected to the appliance

When the relay is activated by the Arduino, the **COM and NO terminals connect**, completing the circuit and turning the appliance **ON**.

5. Overall Circuit Operation

1. The PIR sensor continuously monitors the area.
2. When motion is detected, it sends a signal to the Arduino.
3. The Arduino processes this signal.
4. The Arduino activates the relay and LED.
5. The relay turns ON the connected appliance.
6. When no motion is detected for some time, Arduino turns the relay OFF and the appliance stops.


UNO

Arduino Programming

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

const int PIR_SENSOR = 2;
const int ROOM_LIGHT = 8;
const int OUTSIDE_LIGHT = 9;

LiquidCrystal_I2C lcd(0x27,16,2);

void setup(){
 Serial.begin(9600);
 pinMode(PIR_SENSOR,INPUT);
 pinMode(ROOM_LIGHT,OUTPUT);
 pinMode(OUTSIDE_LIGHT,OUTPUT);
 lcd.init();
 lcd.backlight();
}

void loop(){
 int motionDetected = digitalRead(PIR_SENSOR);

 if(motionDetected == HIGH){
  digitalWrite(ROOM_LIGHT,HIGH);
  digitalWrite(OUTSIDE_LIGHT,LOW);
 }
 else{
  digitalWrite(ROOM_LIGHT,LOW);
  digitalWrite(OUTSIDE_LIGHT,HIGH);
 }

 delay(500);
}
```

Results & Observations

After designing and implementing the **Arduino-Based Energy Optimization System with PIR Motion Control**, the system was tested under different conditions to observe its performance. The results show that the system successfully detects human motion and automatically controls the lighting system, which helps in reducing unnecessary energy consumption.

1. Motion Detection Performance

During testing, the **Passive Infrared Sensor** successfully detected human movement within its sensing range. When a person entered the detection area, the sensor sent a signal to the **Arduino Uno**, which immediately activated the LED or light connected to the system. This confirmed that the motion detection system was functioning properly.

2. Automatic Control of Lighting

The system automatically turned ON the light when motion was detected and turned it OFF when no motion was present. This automatic control reduced the need for manual switching and prevented electricity wastage when the room was empty.

3. LCD Display Monitoring

The **LiquidCrystal I2C LCD 16x2** successfully displayed system messages such as:

- Motion Detected
- Light ON
- No Motion
- Light OFF

It also displayed the **total time the light remained ON and the estimated energy consumption**, which allowed users to monitor energy usage easily.

4. Energy Consumption Calculation

The system calculated energy usage based on the duration for which the light remained ON. By using the formula:

Energy = Power × Time

the program estimated the energy consumed by the light (assuming a 10W bulb). The results showed that when motion was limited, the system significantly reduced the total operating time of the light, thereby saving energy.

5. System Stability

The system worked continuously without major errors. The sensor responded quickly to motion, and the Arduino processed the signals efficiently. The delay used in the program ensured stable readings and prevented unnecessary flickering of the display.

Advantages

1. **Energy Saving**
   The system automatically turns lights ON and OFF based on human presence, which helps reduce unnecessary electricity consumption.
2. **Automatic Operation**
   No manual switching is required because the system detects motion using a **Passive Infrared Sensor** and controls devices automatically.
3. **Low Cost System**
   The project uses affordable components like the **Arduino Uno**, PIR sensor, and relay module, making it economical to implement.
4. **Easy Installation and Maintenance**
   The system is simple to build and can be installed in different locations without complex wiring.
5. **Energy Monitoring**
   The **LiquidCrystal I2C LCD 16x2** displays energy consumption and operating time, allowing users to monitor electricity usage.
6. **Reduces Human Effort**
   Since the system works automatically, it eliminates the need for people to manually turn appliances ON or OFF.

Applications

1. **Classrooms and Educational Institutions**
   The system can automatically control lights in classrooms when students or teachers enter or leave the room.
2. **Smart Homes**
   It can be used in homes to control lighting systems efficiently and reduce electricity bills.
3. **Offices and Workspaces**
   In offices, the system can automatically manage lighting in meeting rooms, cabins, and corridors.
4. **Hospitals and Public Buildings**
   Motion-based lighting can be used in hospital corridors, waiting areas, and restrooms to save energy.
5. **Parking Areas and Staircases**
   The system can control lights in parking areas or staircases where lights are required only when someone passes through.
6. **Libraries and Laboratories**
   It helps maintain efficient energy use in study areas where lighting is needed only when people are present.

Conclusion

The **Arduino-Based Energy Optimization System with PIR Motion Control** was successfully designed and implemented to reduce unnecessary electricity consumption. The system uses a **Passive Infrared Sensor** to detect human movement and a **Arduino Uno** microcontroller to automatically control the lighting system. When motion is detected, the system turns the light ON, and when no motion is detected for a certain period, the light automatically turns OFF.

The system also displays important information such as motion status, operating time, and estimated energy consumption on the **LiquidCrystal I2C LCD 16x2**, allowing users to monitor energy usage easily. The results obtained during testing show that the system works efficiently and responds quickly to changes in motion.

This project demonstrates that simple automation technology can play an important role in **energy conservation and smart energy management**. The system is low-cost, easy to implement, and can be applied in classrooms, homes, offices, and public buildings to reduce electricity wastage.

Overall, the project provides a **practical and effective solution for improving energy efficiency**, and it highlights the potential of using sensor-based automation systems for sustainable energy usage in the future.

Bibliography

1. FOSSEE Project, Indian Institute of Technology Bombay.
   FOSSEE Arduino and Open Source Electronics Learning Resources.
   Available at: https://fossee.in
2. Arduino.
   Arduino Official Documentation and Tutorials.
   Available at: https://www.arduino.cc
3. FOSSEE Project, Indian Institute of Technology Bombay.
   Open-Source Hardware and Embedded Systems Learning Materials.
   Available at: https://fossee.in/embedded-systems
4. Boylestad, R., & Nashelsky, L.
   Electronic Devices and Circuit Theory. Pearson Education.
5. Mazidi, M. A., Mazidi, J. G., & McKinlay, R.
   The 8051 Microcontroller and Embedded Systems. Pearson.